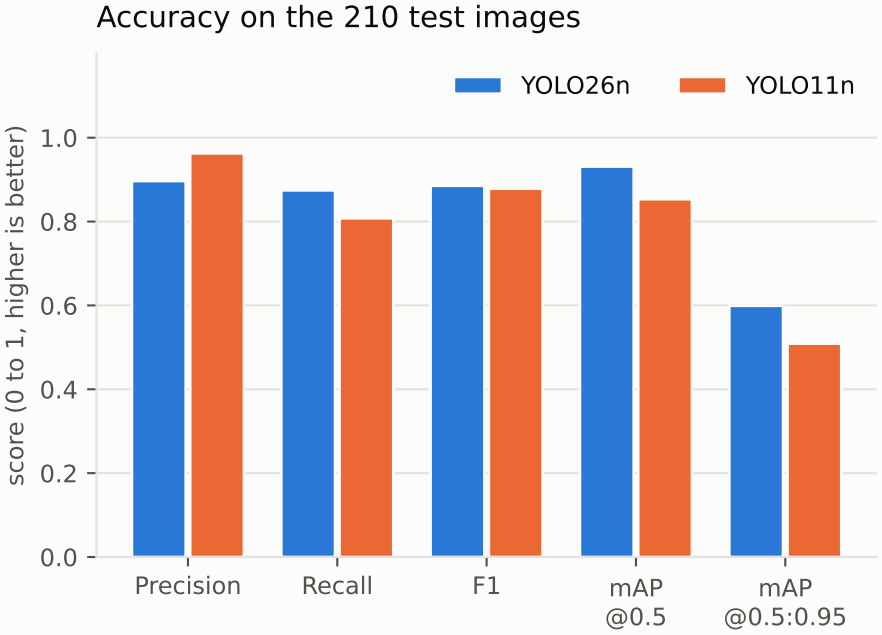
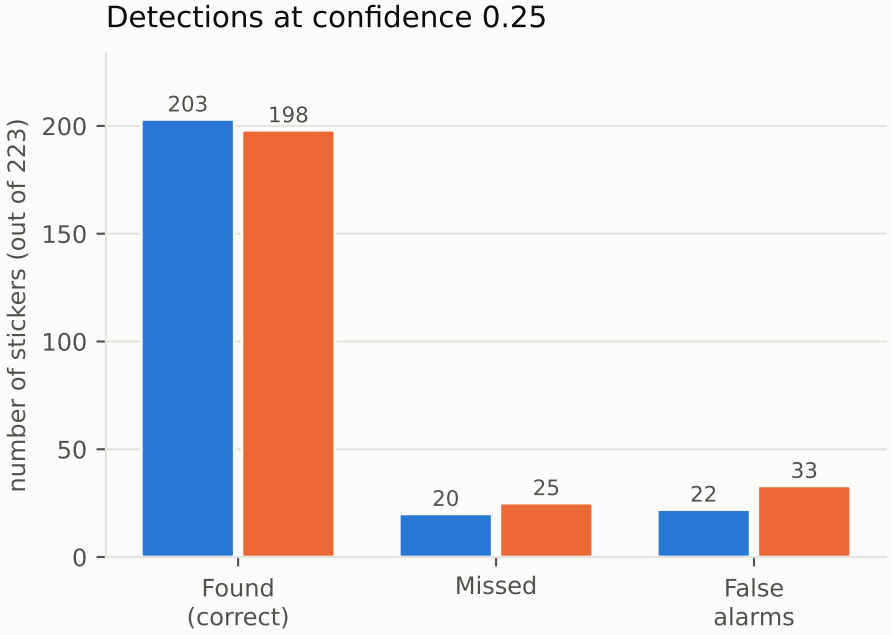
# YOLO26n vs YOLO11n

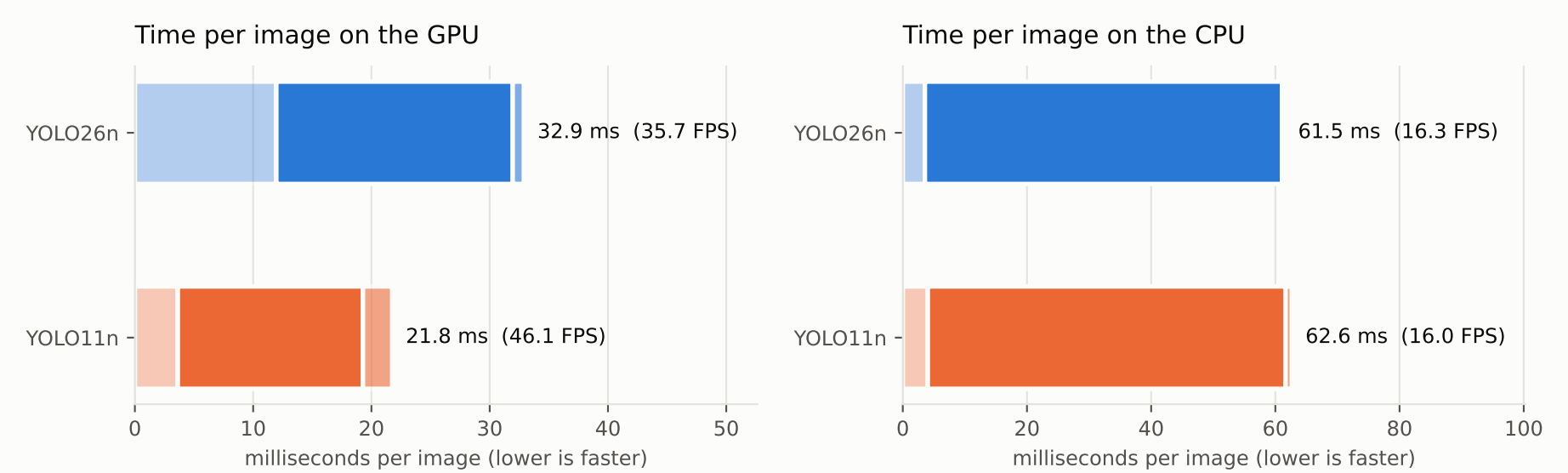
## PSU Vehicle Sticker Detection Benchmark

November 11, 2026

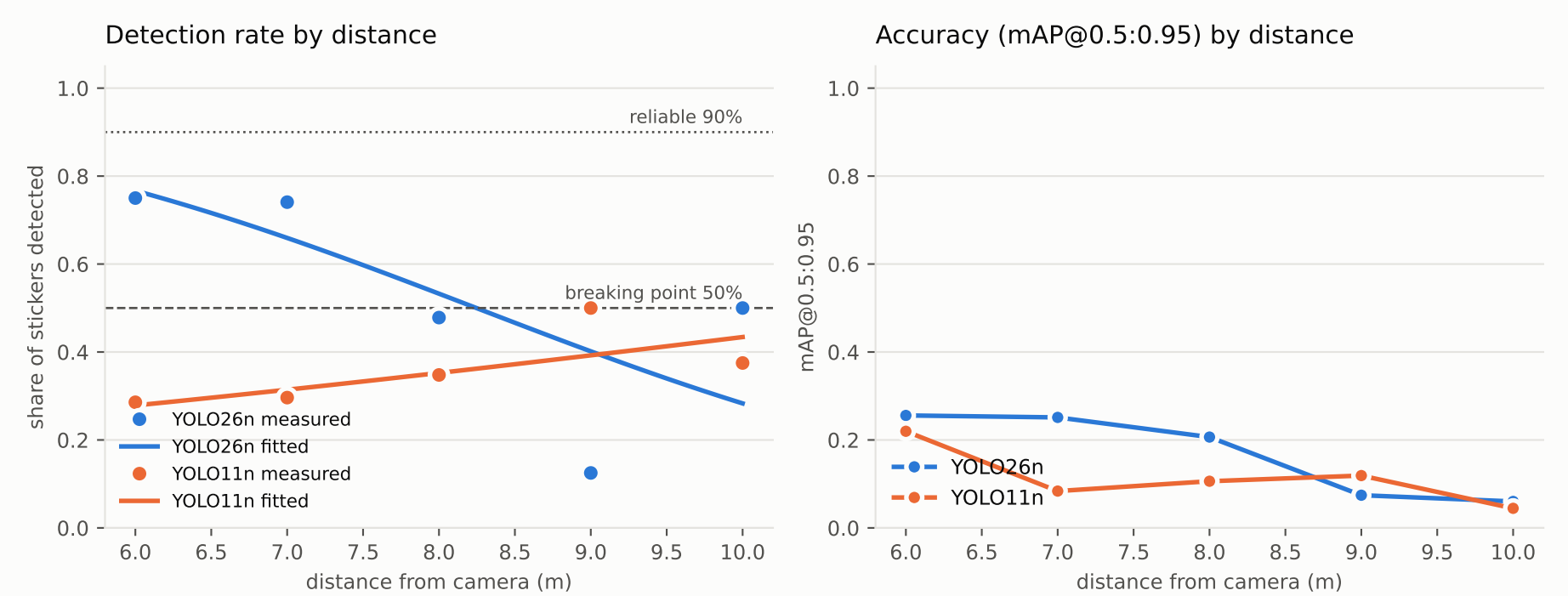
| Measure | YOLO26n | YOLO11n | Better |
|---|---|---|---|
| Accuracy: mAP@0.5:0.95 | 0.599 | 0.509 | YOLO26n |
| Accuracy: mAP@0.5 | 0.931 | 0.853 | YOLO26n |
| Precision | 0.897 | 0.963 | YOLO11n |
| Recall | 0.874 | 0.808 | YOLO26n |
| Speed on GPU (FPS) | 35.7 | 46.1 | About the same |
| Speed on CPU (FPS) | 16.3 | 16.0 | About the same |
| Distance: still detected up to (50%) | 8.2 m | — | YOLO26n found more stickers (McNe |
| Motion blur: breaks at (50%) | 64 px | 71 px | YOLO11n |
| Occlusion: breaks at (50%) | 46% covered | 44% covered | YOLO26n |
| Low light: breaks at (50%) | gamma 3.83 | gamma 4.10 | YOLO11n |
| Glare: breaks at (50%) | gain 3.27x | gain 3.16x | No clear difference (YOLO26n slightl |

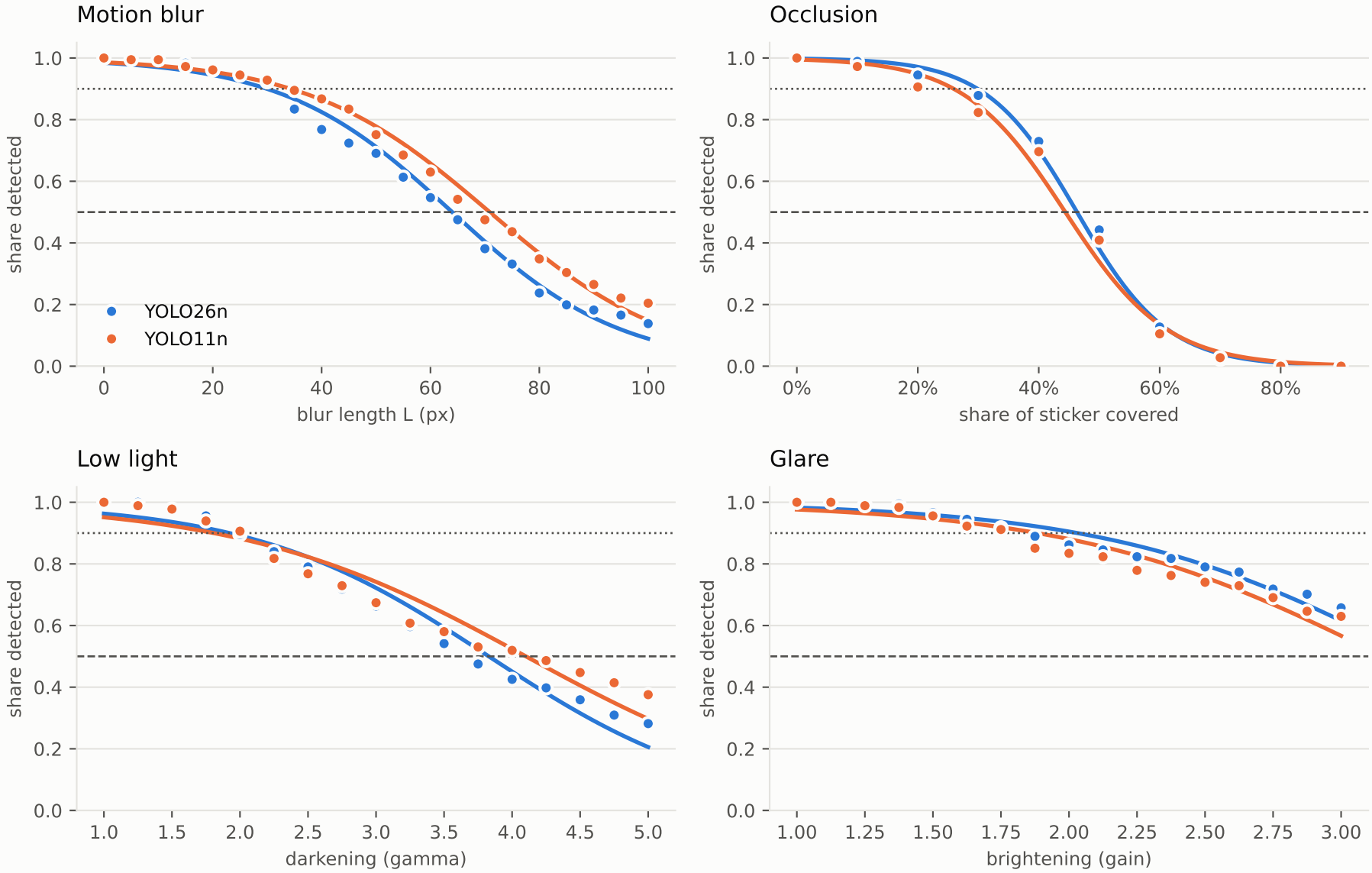

Accuracy on the 210 test images
YOLO26n
YOLO11n
score (0 to 1, higher is better)
1.0
0.8
0.6
0.4
0.2
0.0
Precision
Recall
F1
mAP @0.5
mAP @0.5:0.95


Detections at confidence 0.25
number of stickers (out of 223)
200
150
100
50
0
203
198
20
25
22
33
Found (correct)
Missed
False alarms

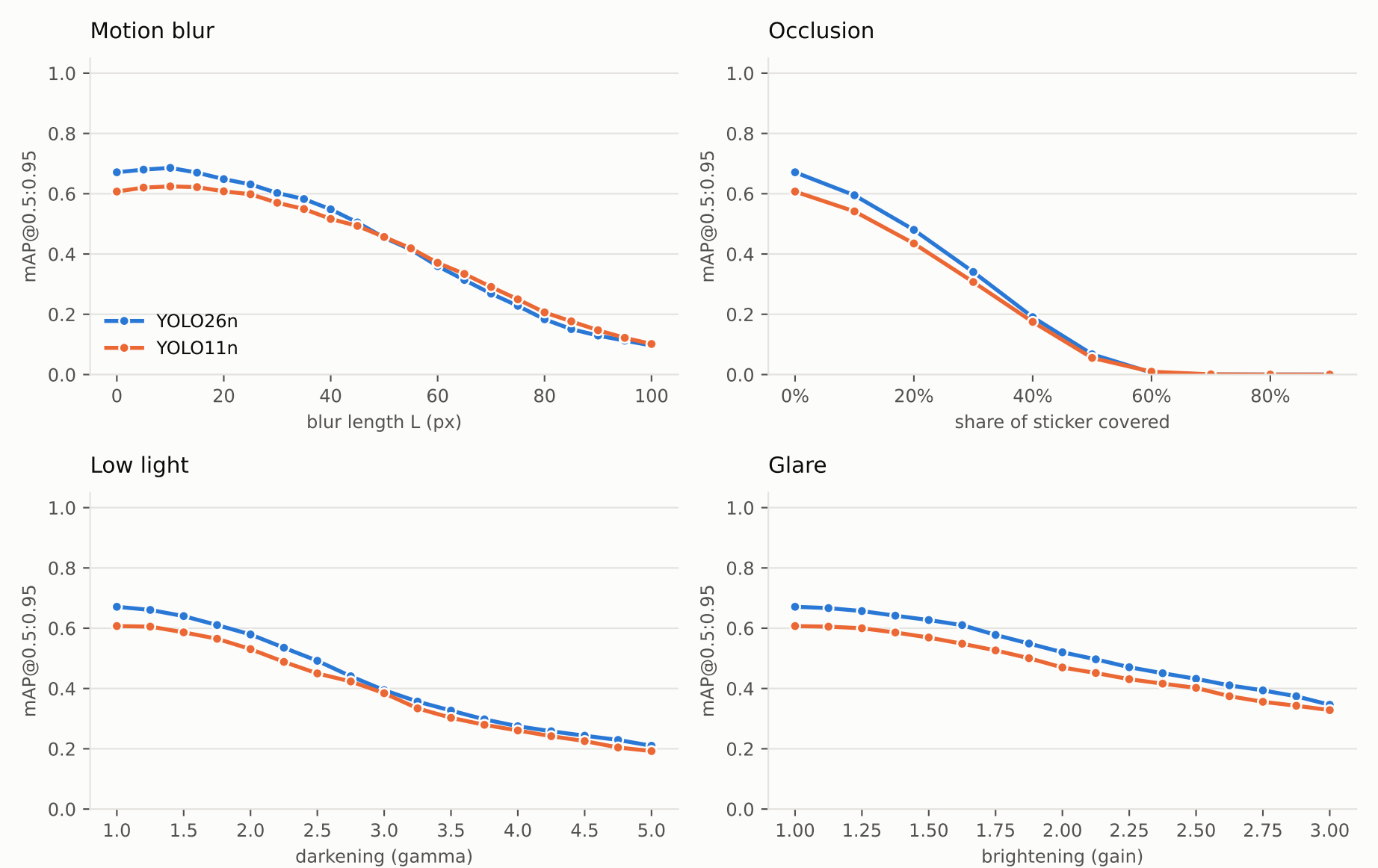



Bar shades, light to dark to medium: prepare image, model, cleanup (NMS). YOLO26n's cleanup is shorter because it is NMS-free.

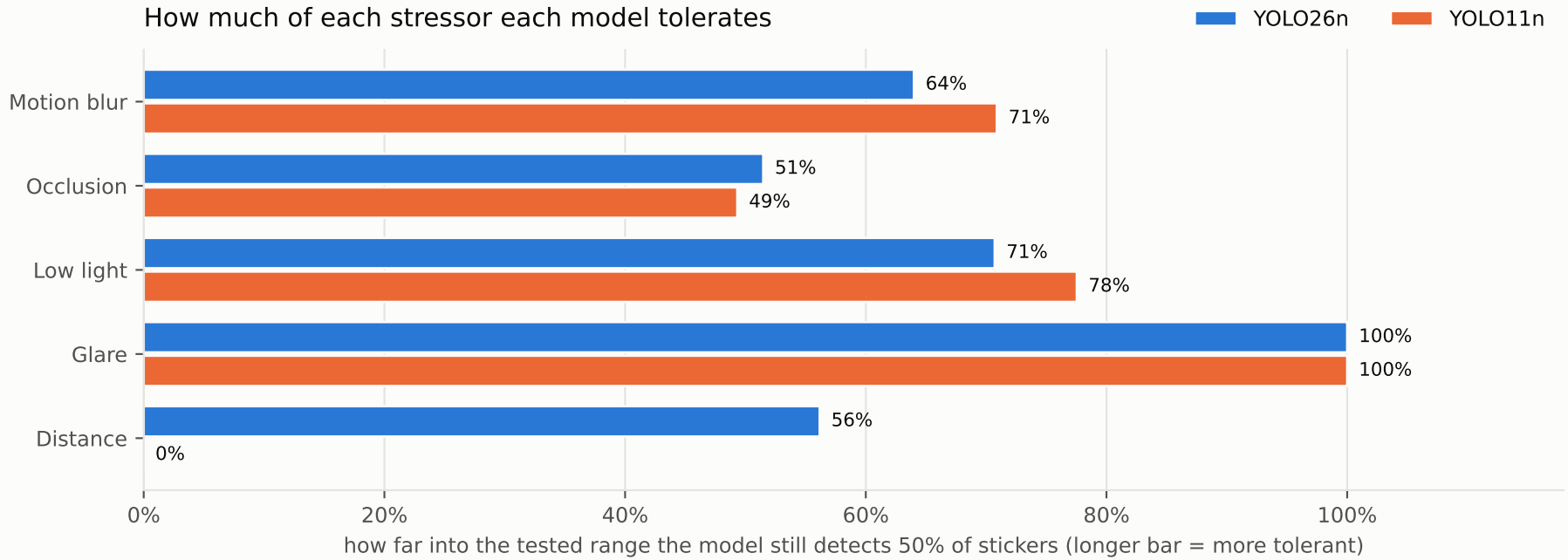

Detection rate by distance
share of stickers detected
1.0
0.8
0.6
0.4
0.2
0.0
reliable 90%
breaking point 50%
YOLO26n measured
YOLO26n fitted
YOLO11n measured
YOLO11n fitted
6.0
6.5
7.0
7.5
8.0
8.5
9.0
9.5
10.0
distance from camera (m)
Accuracy (mAP@0.5:0.95) by distance
mAP@0.5:0.95
YOLO26n
YOLO11n
distance from camera (m)




Dots = measured, lines = fitted curve. Dashed line = breaking point (50% detected); dotted line = reliable detection (90%). Further right = harder.




Further right = harder.


How much of each stressor each model tolerates
YOLO26n
YOLO11n
Motion blur
64%
71%
Occlusion
51%
49%
Low light
71%
78%
Glare
100%
100%
Distance
56%
0%
0%
20%
40%
60%
80%
100%
how far into the tested range the model still detects 50% of stickers (longer bar = more tolerant)